



Figure 1: Welcome screen



Figure 2: Hosting static Web content from a public directory

parsing the body of the content that comes from client and request routing are performed by middleware.

Features of Express.js

Using Express, you can host static Web pages and also highly scalable Web applications such as blogging sites and e-commerce sites. Express, by itself, does not impose any restrictions on how you use the framework or build your project structure. It offers complete flexibility for you to define. Express retains Node's philosophy in terms of simplicity and maintains the same functions as *connect* middleware.

The Jade Template engine from the developers of the Express framework extends the capabilities towards the HTML pages and CSS. Jade has programming constructs such as looping, variables, etc, which make for the generation of Web pages, but an introduction to Jade is not within the scope of this article.

Installation and set-up

Installing Express is similar to that of any other Node package. To install Express-dependent modules and also the binary files required, run the following command:

```
$ npm install -g express
```

The *-g* option installs Express in the global mode.

To check the installed version of Express, run the command given below:

```
$ express -v
3.4.3
```

Developing a Web app

Let's now learn how to use Express to host a website with static HTML files and also develop a Web server to view pictures from a folder on the server. Though the photo viewer project in this article only displays thumbnails with a border, you can extend the project to feature a richer photo viewer.

Let's instruct Express to create a project named 'photoviewer':

```
$ express photoviewer

create : photoviewer
create : photoviewer/package.json
create : photoviewer/app.js
create : photoviewer/public
create : photoviewer/routes
create : photoviewer/routes/index.js
create : photoviewer/routes/user.js
create : photoviewer/views
create : photoviewer/views/layout.jade
create : photoviewer/views/index.jade
create : photoviewer/public/stylesheets
create : photoviewer/public/stylesheets/style.css
create : photoviewer/public/javascripts
create : photoviewer/public/images
```

Install dependencies, as shown below:

```
$ cd photoviewer && npm install
```

Run the app, as follows:

```
$ node app

$ cd photoviewer
$ sudo npm install
Password:
<output of dependent modules installed>
$ node app
Express server listening on port 3000
```

Open the browser and go to <http://localhost:3000>

Command line options (*less*, *hogan* and *ejs*)

Express has many command line options to create project templates. These are: session support, different template engines for HTML and CSS. The template engines for HTML are Jade, EJS and *jshtml*. For stylesheets, you can choose *less* or *stylus*. Jade is the default engine in Express.